

Muhammad Azani Hasibuan

M.T.I. · Lecturer in Information Systems · PhD Candidate in Computer Science

Lecturer in Information Systems at Telkom University and PhD candidate at the University of Western Australia. I combine more than fifteen years of experience in university teaching, learning technology leadership and IT consulting. My research applies learning analytics, data mining and process mining to understand and support students' self-regulated learning in online and collaborative environments.

Web: muhammadazani.github.io · **LinkedIn:** linkedin.com/in/muhammadazani · **GitHub:** github.com/muhammadazani

Research interests: Learning analytics · Data mining & business intelligence · Online and collaborative learning · Human-computer interaction · Self-regulated learning · Process mining

Education

- 2020–present** **PhD in Computer Science (Computer Science Education)**, in progress
The University of Western Australia, Perth · Department of Computer Science and Software Engineering
Thesis: A systematic framework for context- and domain-driven event abstraction in multi-modal data within online collaborative learning environments.
- 2009–2011** **Master of Information Systems**
Universitas Indonesia, Faculty of Computer Science · GPA 3.61/4.00
- 2003–2008** **Bachelor of Informatics Engineering**
Universitas Indonesia, Faculty of Computer Science

Academic employment

- 2014–present** **Faculty Member, Department of Information Systems**
Telkom University, Bandung
Teaching programming, databases, data mining, business intelligence and project management to undergraduate Information Systems students, and supervising undergraduate research projects.
- 2018–2020** **Head of Learning Technology, Center for Teaching & Learning Excellence (CTLE)**
Telkom University, Bandung
- Led the university-wide adoption of blended learning across all faculties.
 - Helped faculty members design and implement blended-learning courses.
 - Managed the development of more than 500 online courses for all schools.
 - Evaluated the implementation of blended learning and e-learning.
- 2011–2013** **Earlier teaching**
Lecturer, Politeknik Negeri Jakarta (2013) · Part-time Lecturer, STMIK Inti Indonesia (2011–2012) · Guest Lecturer, Sekolah Tinggi Manajemen Telkom (2011)
- Courses: Enterprise resource planning · Knowledge management · Principles of e-business · Web multimedia · Business process management

Teaching

- 2020** **The University of Western Australia**
Lab facilitator, Relational Database Management Systems
- 2014–2019** **Telkom University, undergraduate Information Systems**

Course	Semesters
Introduction to Computer Algorithms (Java)	2014-2, 2015-2

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Database Systems	2014-2, 2015-2
Data Structures and Algorithms (Java)	2015-1, 2016-1, 2017-1
Human-Computer Interaction	2015-1
Project Management for Information Systems	2016-1, 2017-1, 2018-1, 2019-1
Mobile Application Development (Android)	2016-2, 2017-2
Capstone: Software Engineering Project	2016-2, 2017-2, 2018-2, 2019-2
Data Warehouse & Business Intelligence (Tableau)	2018-1, 2019-1
Data Mining (Python)	2018-2, 2019-2

Current courses are listed on the [Teaching](#) page.

Industry experience

2009–2013

Senior IT Consultant & Solution Architect

PT Totalindo Rekayasa Telematika, Jakarta

Designed and delivered business-process, technology-infrastructure and IT-governance strategies for public- and private-sector clients. Key projects:

- **Ministry of Home Affairs, e-KTP national ID programme (2011–2012):** designed the helpdesk business processes and led the agile customisation, rollout and agent training of the helpdesk management system.
- **PUSLIA, West Java Regional Revenue Office (2012–2013):** designed IT service standard operating procedures and implemented an open-source ticketing system.
- **Perum Bulog (2011–2012):** developed the IT/IS blueprint and roadmap, then led the ERP initiation: requirements, technical specification, cost and schedule estimates, and terms of reference.
- **Baitul Maal Hidayatullah (2010–2011):** led the implementation of an open-source ERP system.

2008–2009

Data Engineer

PT Solusi247, Jakarta

Built data warehouse and business intelligence solutions for Indonesian telecommunications companies, including Indosat, NTS Axis (with IBM Indonesia) and Bakrie Telecom.

Research experience

2016–2018

Enterprise Data Management Lab, Telkom University

- Analysed how GRIT, self-efficacy and gamification relate to student performance in blended learning (2017–2018).
- Applied Apriori association rules to web-usage logs and GRIT scores to identify factors that determine student outcomes.
- Built a Pentaho ETL pipeline for two million rows of student data, leading to the paper *Soft Set Approach for Clustering Graduated Dataset* (Springer, 2017).

Publications

Twenty-four publications since 2013, in venues including *Sustainability*, *Journal of Computers*, Springer and IEEE conferences, and ASCILITE. Selected work:

- Hasibuan, M. A., Reynolds, M., Male, S., & Hassan, G. M. (2023). Role of theory in analysing the dynamic of self-regulated learning process based on students' event logs data: A scoping review. *ASCILITE 2023: People, Partnerships and Pedagogies*, 126–136.

- Hasibuan, M. A., Andreswari, R., Lubis, M., Perdana, F., & Fakhurroja, H. (2023). Classification algorithm for predicting the graduation level: Case study of e-learning in Telkom University. In *Intelligent Sustainable Systems* (Vol. 1). Springer.
- Lubis, M., Hasibuan, M. A., & Andreswari, R. (2022). Satisfaction measurement in the blended learning system of the university: The Literacy Mediated-Discourses (LM-D) framework. *Sustainability*, 14(19), 12929.
- Hasibuan, M. A., Reynolds, M., Male, S., Hassan, G. M., & Kusumasari, T. F. (2021). Assessing a measurement model of self-regulated learning in an online collaborative learning environment. *REES AAEE 2021 Conference*.
- Putri, D. Y., Andreswari, R., & Hasibuan, M. A. (2018). Analysis of students graduation target based on academic data record using C4.5 algorithm. Case study: Information Systems students of Telkom University. *6th International Conference on Cyber and IT Service Management (CITSM)*, 1–6. IEEE.
- Dwiandriani, F., Kusumasari, T. F., & Hasibuan, M. A. (2017). Fingerprint clustering algorithm for data profiling using Pentaho Data Integration. *2nd International Conference on Information Technology, Information Systems and Electrical Engineering (ICITISEE)*, 359–363. IEEE.
- Saedudin, R. R., Kasim, S. B., Mahdin, H., & Hasibuan, M. A. (2017). Soft set approach for clustering graduated dataset. In *Recent Advances on Soft Computing and Data Mining* (pp. 631–637). Springer.
- Hidayanto, A. N., Hasibuan, M. A., Handayani, P. W., & Sucahyo, Y. G. (2013). Framework for measuring ERP implementation readiness in small and medium enterprise (SME): A case study in software developer company. *Journal of Computers*, 8(7), 1777–1782.

[Full publication list →](#)

Skills

Programming & data

Python (pandas, NumPy, scikit-learn) R Java
SQL / SQLite Oracle PL/SQL

Research methods

Learning analytics Data mining Process mining
Machine learning Text mining

Analytics & BI platforms

Tableau Pentaho Data Integration IBM Cognos
IBM DataStage

IS/IT management

Project management IT governance (COBIT, ITIL)
Business process management Enterprise resource planning
Scrum & agile

Language & professional development

Language

IELTS overall band 7.0

2007–2012

Professional Scrum Foundation, Maranatha Christian University (2012)
Enterprise Architecture with MEGA Suite, Institut Teknologi Telkom (2012)
ISO 9001:2008, PT Totalindo Rekayasa Telematika (2012)
Business Intelligence with IBM Cognos, PT Solusi247 (2008)
Probabilistic Robotics short course, Universitas Indonesia (2007)